

SENTINELROOM: ADAPTIVE INTERROGATION ENVIRONMENT OVER ISOLATED MILITARY NETWORKS

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



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1 ABSTRACT

Military area networks are critical infrastructures that demand high resilience and performance in dynamic operational environments. The advent of Software-Defined Networking (SDN) presents a transformative approach to managing these networks by enabling centralized control and programmability. However, a significant challenge lies in the comprehensive and flexible testing of these networks under a wide array of realistic scenarios, from routine operations to sophisticated cyber attacks. Current testing methodologies are often rigid, complex, and fail to fully leverage SDN's programmable nature for dynamic scenario generation. This research proposes a novel methodology and a corresponding simulation framework that utilizes SDN technologies to facilitate the programmable and automated testing of performance and resilience in military networks. The core of this work is the development of a high-level interface that allows for the easy definition and execution of diverse test cases, including network failures, cyber attacks, and dynamic traffic conditions. The proposed framework integrates network emulation tools with an SDN controller, providing a realistic environment to evaluate key performance metrics such as throughput and latency, alongside resilience metrics like network recovery time. The expected outcome is a scalable and extensible simulation platform that significantly simplifies and accelerates the testing process, enabling a more thorough evaluation of network robustness and adaptability. This will provide military network operators and planners with a powerful tool to better design, deploy, and manage resilient communication systems for mission-critical operations.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
SDNs	Software Defined Networks
NFV	Network Function Virtualization
EMANE	Extendable Mobile Ad-hoc Network Emulator
ARMS	Applied Resilience for Mission Systems
CORE	Common Open Research Emulator
TNM	Tactical Network Modeller
ARM	Applied Resilience for Mission Systems
LSTM	Long Short Term Memory

2 INTRODUCTION

During or before military operations the safety of the military network can directly affect the assets that the military are responsible of which include human lives whom they have to protect to the best of their ability. Military area networks are thus highly critical infrastructure, high amount availability, resilience, performance and adaptability is expected in a dynamic and hostile environment. In a study published on IEEEExplore this has been clearly stated, saying that robustness and resilience of communication are most crucial for a military network.[1] Software defined networks not only offer a solution to improve the overall survivability or functionality of the network but also provides the ability for novel solutions such as emulation of attack scenarios to train operators or simply predict and prepare for adversarial counter measures. This is possible using the separation of the network into control and data planes for more flexible and easily controllable network architecture which is expected change or adapt according to a dynamic environment. This is crucial for tactical scenarios due to the unique nature of cyber attacks which are generally more sophisticated and demand an extended process which involves collecting intelligence data, testing and evaluating attacks against the deployment before real life deployment.

Currently methods used for testing military networks are often related to live field exercises which are expensive, harder to manipulate replicate or simulate environments that can be complex and lack the flexibility to setup the required testbed.[2] [3] Some of the existing solutions are simply limited to simple network simulation rather than addressing the complex needs of such a network or rather has not gone beyond a simple proof of concept. [3] This research proposes a comprehensive framework to assist the proper and extended methodology in order for a successful, effective and efficient solution for addressing these problems related to evaluation of military networks. It is expected to create a programmable environment where various network scenarios can be simulated including cyber-attacks and military specific devices which can be easily defined and deployed in order test the performance and resilience of the network. The following solution is intended to improve the cost efficiency, ease of creating and deploying scenarios related to collected intelligence data and speed of implementation and deployment with the ability to adapt to a dynamic environment with turbulent threats and network environment.

2.1 Background & Literature Survey:

The fight generation of telecommunication standards which are tied to technologies such as software Defined Networks (SDNs) and Network Function Virtualization (NFV) has drawn attention to the said technologies in the recent years [4]. SDN's centralized control architecture provides more visibility to the network which makes it easier to manage and control, resulting the increase of efficiency in routing, resource allocation and policy enforcements. This network management architecture enables the network to be programmable, making it suitable for instances where high adaptability and dynamic nature of both network and network infrastructure is observed or expected by decoupling the network routing process (Control Plane) and the network packet(Data Plane). This idea mainly originated from the public switch telephone networks where control and data plane were segregated before it was applied for the contemporary networks [5].

The use of SDNs for performance and security enhancement and evaluation is cited in several studies, they have shown that it can improve and optimize functions such as routing and load balancing. From a security perspective, the centralized architecture can act as a single point of control for policy enforcements, which allows efficient and robust implementation of security measures.

Simulation and emulation tools are widely used for evaluation network performance and resilience, tools such as Common Open Research Emulator [6] and Extendable Mobile Ad-hoc Network Emulator (EMANE) are used for realistic network model. Furthermore, specialized testbeds like the Applied Resilience for Mission Systems (ARMS) have been developed for experimenting with and assessing the resilience of mission critical systems.

Simulation and emulation are widely used for evaluating network performance and resilience. Tools like the Common Open Research Emulator (CORE) and the Extendable Mobile Ad-hoc Network Emulator (EMANE) are used to create realistic network models, including detailed radio models for wireless communication. Current frameworks such as the Tactical Network Modeller (TNM) aim to simplify the process of building network model. Applied Resilience for Mission Systems (ARMS) have been developed for experimenting with and assessing resilience of networks.

Simulations for analysis of impact from cyber attacks on networks have also been explored, some research focus on developing machine learning for anomaly and attack detection in SDN environment [8]. Methodologies based on Markov models and Long Short Term Memory (LSTM) networks have been proposed to test the resilience of SDN [9]. Commercial SDN technologies also exists for military network simulation/emulation [6].

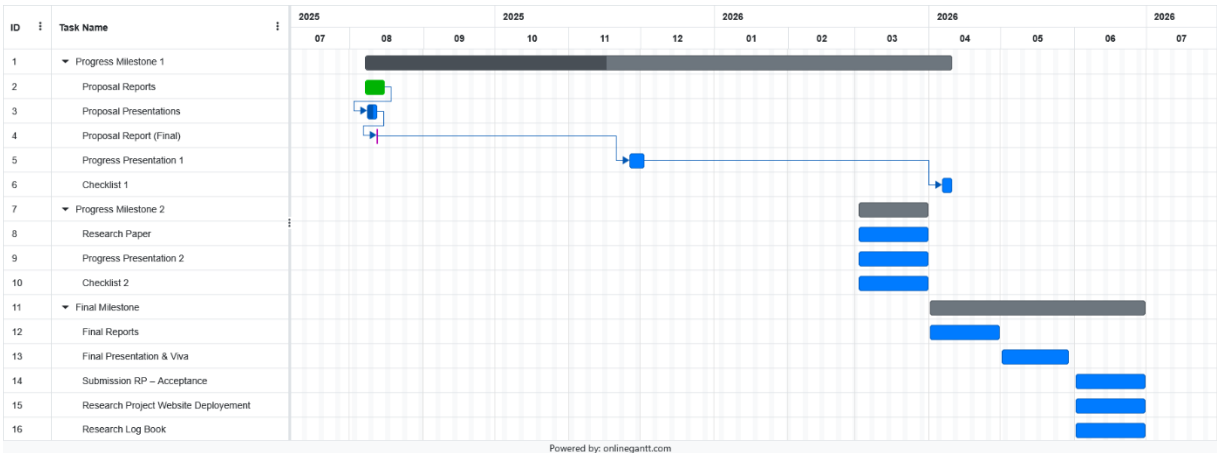
2.2 Research Gap:

Despite the existing body of work, a significant research gap remains in the intersection of SDN, military network simulation, and programmable test case generation for comprehensive performance and resilience analysis. While current simulation tools and testbeds offer valuable capabilities, they often require specialized expertise and significant effort to create and modify test scenarios. There is a lack of a unified methodology and framework specifically designed to provide an easy and programmable way to define and execute a wide range of test cases that can simultaneously evaluate both performance metrics and network resilience under various operational and threat scenarios. Existing solutions tend to focus on specific aspects, such as attack simulation or network modeling, but do not provide an integrated and easily programmable environment for holistic testing.

2.3 Research Problem:

The central research problem is the lack of an efficient and flexible methodology to simulate and evaluate the performance and resilience of military area networks that leverage SDN technologies. The current approaches for testing are often rigid, time consuming, and do not fully exploit the programmability offered by SDN to create dynamic and realistic test scenarios. This makes it challenging to comprehensively assess how a network will behave under a multitude of conditions, ranging from routine operations to sophisticated cyber attacks and unexpected network failures. Therefore, this research aims to answer the question: How can a programmable simulation framework be designed and implemented to enable the rapid and flexible testing of performance and resilience in SDN-based military area networks under dynamic scenarios/environment?

2.4 Research Timeline



3 OBJECTIVES

3.1 Main Objective:

The main objective of this research is to develop a methodology and a solution that utilizes SDN principles to enable an easily programmable framework for performance and resilience testing in military networks.

3.2 Sub-Objectives:

To design a modular and extensible simulation architecture that integrates an SDN controller, network emulation tools, and a programmable test case generation engine.

To develop a high-level, user-friendly programming interface or language for defining a wide range of test scenarios, including network topologies, traffic patterns, device mobility, link failures, and various cyber-attack vectors.

To implement a set of core performance metrics and resilience evaluation modules within the framework, such as throughput, latency, packet loss, network recovery time, and quality of service (QoS) degradation under stress.

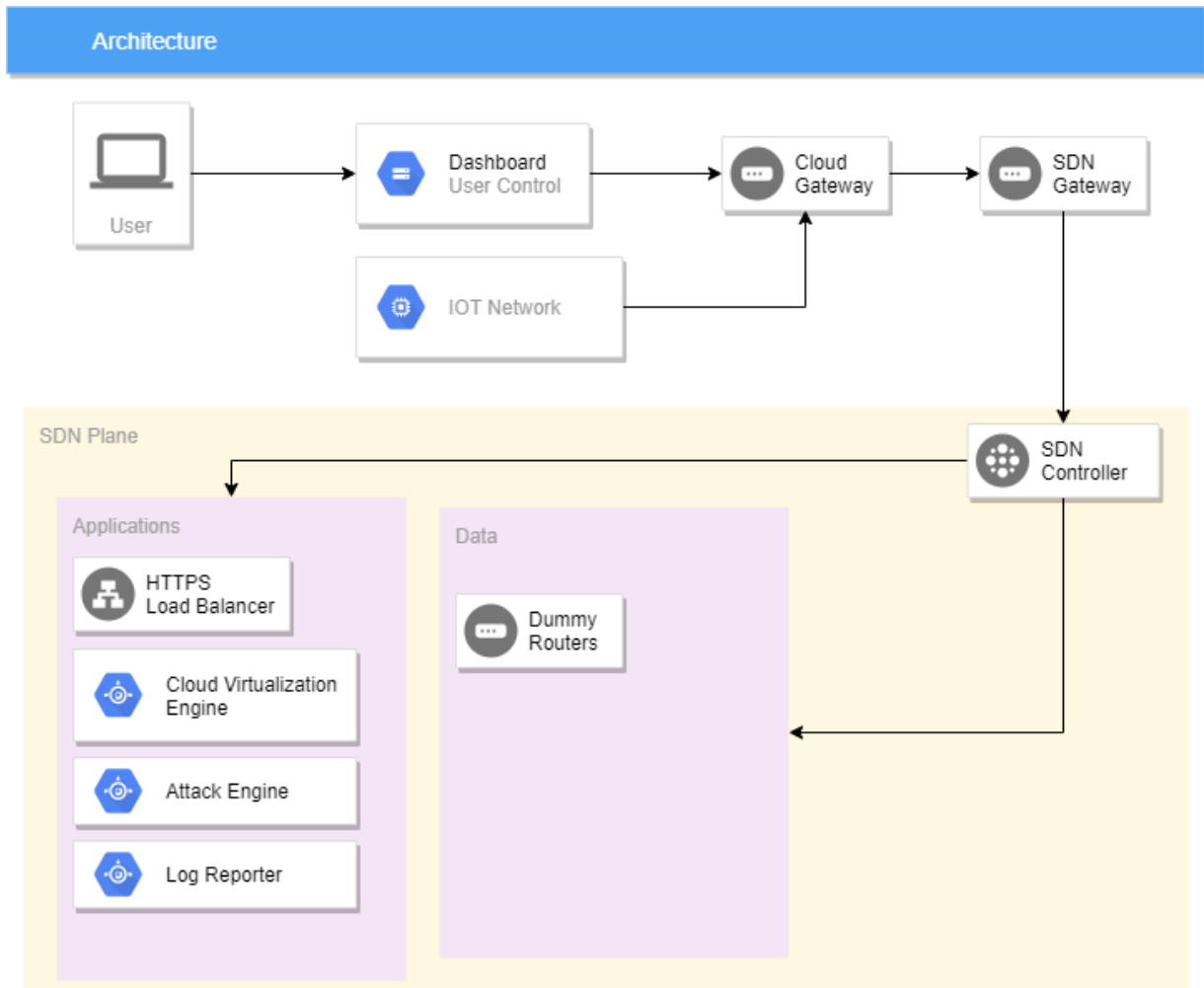
To create a library of predefined, customizable test cases that represent realistic military operational scenarios and common cyber threats.

To validate the proposed framework by simulating a representative military area network and evaluating its performance and resilience against a set of defined test cases.

To analyze the results to demonstrate the effectiveness and flexibility of the programmable testing methodology compared to traditional approaches.

4 METHODOLOGY

4.1 Proposed Solution



A system is proposed in order to orchestrate a virtual network using Network Virtualization Functions available within SDN frameworks. There after a separate system is created in order to create realistic cyber attacks within the system, another system will be responsible of actively monitoring the changes that are occurring to the network. This system is called the Resilience Engine, it is also responsible for implementing resilient protocols and self healing or other flow control rules inorder to simulate a function of a Network Anomaly Prevention system used within a military network context.

4.2 Project requirements

Following covers the main requirements of the the project.

4.2.1 Functional requirement

The system shall allow users to define custom network topologies, including nodes, links, and their characteristics.

The system shall support the simulation of both wired and wireless network environments.

The system shall provide an interface to an SDN controller for managing the simulated network.

The system shall allow users to define and inject various types of network traffic with configurable parameters.

The system shall enable the simulation of dynamic network events, such as node mobility, link failures, and congestion.

The system shall support the introduction of a range of cyber-attacks, including but not limited to Denial of Service (DoS), Distributed Denial of Service (DDoS), and man-in-the-middle attacks.

The system shall collect and log key performance and resilience metrics during the simulation.

The system shall provide a mechanism for visualizing the network topology and the simulation results.

4.2.2 Non Functional Requirements

The framework should provide a user-friendly interface for defining and executing test cases, minimizing the learning curve for network engineers and researchers.

The simulation environment should be able to support networks of varying sizes, from small tactical teams to larger area networks.

The architecture should be modular, allowing for the easy integration of new network protocols, device models, and attack modules.

The simulation should run in a reasonable amount of time to allow for iterative testing and analysis.

The framework must ensure that simulations are reproducible to allow for the verification of results.

4.3 Expected Test Cases

Baseline Performance: Measuring throughput, latency, and packet loss under normal operating conditions with varying traffic loads.

Network Resilience to Failures:

Simulating the failure of a key network node (e.g., a router or a base station) and measuring the time it takes for the network to reconverge and restore connectivity. Simulating the degradation and eventual failure of a communication link to assess the network's ability to reroute traffic.

Resilience to Cyber-Attacks:

- **DoS/DDoS Attack:** Launching a simulated DoS/DDoS attack against a critical network node or service and evaluating the impact on network performance and the effectiveness of mitigation techniques implemented in the SDN controller.

- **Man-in-the-Middle Attack:** Simulating an attack where an adversary intercepts and potentially alters communication between two nodes to test the effectiveness of security protocols.

Mobility Scenarios: Simulating the movement of mobile ad-hoc network (MANET) nodes and evaluating the network's ability to maintain connectivity and QoS for mobile units.

Quality of Service (QoS) Prioritization: Testing the SDN controller's ability to prioritize critical traffic (e.g., voice or video streams) over less important data during periods of network congestion or degradation.

Combined Scenarios: Creating complex scenarios that combine multiple events, such as a network failure occurring during a cyber-attack, to test the overall robustness of the network.

5 BUDGET AND BUDGET JUSTIFICATION

5.1 Equipment

Category	Year 1	Year 2	Total
High-Performance Server	Rs 15,000	0	Rs 15,000
Workstations (2)	Rs 6,000	0	Rs 6,000
Networking Equipment	Rs 2,000	0	Rs 2,000
Subtotal	Rs 23,000	0	Rs 23,000

Justification:

High-Performance Server: A dedicated high-performance server is crucial for running complex and large-scale network simulations. This server will host the SDN controller, the network emulator, and the simulation management software. The specifications will include multiple high-core-count processors, a significant amount of RAM (e.g., 256GB or more), and fast storage to handle the computational demands of the simulations. A budget of Rs. 15,000 is allocated for a suitable server (hiring).

Workstations: Two high-end workstations are required for the PI and the RA to develop the simulation framework, design test cases, and analyze the results. These workstations will be equipped with powerful processors, ample memory, and graphics capabilities for visualization.

Networking Equipment: Basic networking hardware, including switches and cables, will be necessary to create a physical lab environment for testing and validation of certain aspects of the simulation framework.

5.2 Software

Category	Year 1	Year 2	Total
Operating Systems	Rs 1,000	Rs 0	Rs 1,000
Virtualization Software	Rs 800	Rs 800	Rs 1,600
Subtotal	Rs 1,800	Rs 800	Rs 2,600

5.2.1 Justification:

Operating Systems: While many of the tools will run on Linux, licenses for Windows or other operating systems may be required for specific software or testing scenarios.

Virtualization Software: Licenses for virtualization software like VMware may be needed. While academic discounts are sometimes available, recent changes in VMware's licensing suggest that costs could be higher than in the past. This budget line item accounts for potential licensing fees.

Open-Source Software: The core of the simulation framework will be built using open-source tools to minimize costs and maximize flexibility. This includes network simulators like ns-3 or CORE, and SDN controllers such as OpenDaylight or Ryu. There are no direct license costs for this software, but the

personnel budget covers the time and expertise required for their setup, configuration, and customization.

5.3 Other Direct Costs

Category	Year 1	Year 2	Total
Travel	Rs 3,000	Rs 3,000	Rs 6,000
Publications	Rs 1,500	Rs 1,500	Rs 3,000
Subtotal	Rs 4,500	Rs 4,500	Rs 9,000

Justification:

Travel: This budget will support the PI and RA to travel to one major academic conference each year to present research findings, interact with other researchers in the field, and stay current with the latest advancements.

Publications: Funds are allocated to cover the costs associated with publishing research articles in peer-reviewed journals and conference proceedings, including any open-access fees.

5.4 Indirect Costs

Category	Year 1	Year 2	Total
Indirect Costs (40% of Direct Costs)	Rs 39,280	Rs 27,200	Rs 66,480
Subtotal	Rs 39,280	Rs 27,200	Rs 66,480

5.4.1 Justification:

Indirect costs are calculated at a rate of 40% of the total direct costs (Personnel, Equipment, Software, and Other Direct Costs). This covers administrative and facility costs provided by the research institution that are not directly associated with the project but are necessary for its execution.

6 BUDGET AND BUDGET JUSTIFICATION

No dedicated hardware purchases are required for this contribution, as the entire network is simulated on general-purpose computing equipment. The following resources will be used:

Hardware: Personal computer or lab workstation (existing resources). No additional hardware costs.

Software: All software components are open-source or free for academic use. This includes Linux/Ubuntu OS, Mininet, NS-3, OpenVPN, Python, and any chosen honeypot software.

Licenses: No commercial licenses are needed. All tools (OpenVPN, Wireshark for monitoring, etc.) are available at no cost.

In summary, the budgetary requirements are minimal: since no hardware is needed and all tools are free, the cost impact is effectively zero for this contribution.

This ensures that the contribution can be developed, simulated, and validated at zero additional financial cost.

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